



FACULTY OF INFORMATION TECHNOLOGY

SELF-REFLECTIVE REPORT

Object-Oriented Programming Fundamentals in Java

Assignment 1

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1. Introduction

This report reflects my learning experience throughout the Object-Oriented Programming tutorials using Java. The work in my GitHub repository shows a progression from simple Java programs to classes, inheritance, polymorphism, file handling, and a Java GUI application.

The tutorials also gave me practical experience with Visual Studio Code, Git, and GitHub. Managing the work as one repository helped me understand that programming is not only about writing code, but also about organizing files, checking errors, and keeping a project properly documented.

2. Knowledge Gained from Each Tutorial

Tutorial 1

Tutorial 1 introduced me to the basic structure of Java programs. My repository contains HelloWorld.java and StudentGrade.java, which gave me practice creating, compiling, and running simple Java programs.

Tutorial 2

Tutorial 2 continued with basic Java programming through Main.java and Student.java. This helped me become more comfortable with Java syntax and working with more than one source file.

Tutorial 3-4

Tutorials 3 and 4 were an important step into Object-Oriented Programming. The repository contains Person.java, Student.java, Lecturer.java, and Main.java. Working with these classes helped me understand how different objects and classes can be organized within one program.

Tutorial 5

Tutorial 5 continued the use of classes and objects through Student.java and Main.java. The documentation.txt file also shows that the tutorial involved supporting project documentation. This helped me understand the importance of keeping both code and project information organized.

Tutorial 6

Tutorial 6 focused on inheritance and relationships between classes. The repository contains Employee.java, Lecturer.java, and Main.java. Working with these classes helped me understand how one class can build on another and how inheritance can reduce repeated code.

Tutorial 7

Tutorial 7 developed my understanding of inheritance and polymorphism through Appliances.java, Refrigerator.java, WashingMachine.java, and Main.java. This showed me how related classes can share common features while still having their own behavior.

Tutorial 8-9

Tutorials 8 and 9 focused on working with files. The repository contains Main.java and task.txt. This gave me practical experience with using external text data instead of keeping all information inside the Java source code.

Tutorial 10

Tutorial 10 was the most complete application in the repository. It contains Questions.java and QuizBattleGUI.java and represents a Java Quiz Battle GUI application. This tutorial gave me experience working with a graphical interface and combining several parts of a Java project.

3. Challenges Encountered

One of the main challenges was understanding how multiple Java classes work together. As the tutorials became more advanced, it became important to keep class names, file names, and relationships between classes correct.

A specific issue occurred in Tutorial 6 when I received a duplicate class: Main error. The problem was related to an extra Main.java file in the tutorial folder. I also needed some practice to become comfortable with Git commands such as git add, git commit, git push, and git status.

4. How I Overcame the Challenges

I learned to read the compiler error carefully and check the project files before changing the code. For the Tutorial 6 problem, I checked the folder, identified the extra Main.java file, removed the incorrect file, and tested the program again.

For GitHub, I used git status to check my changes before committing them. I then used git add, git commit, and git push to update the repository. This made the process clearer and helped me understand how version control works in practice.

5. Improvements in My Java Skills

My Java skills improved from writing simple programs to working with several related classes and a GUI application. I became more comfortable reading code, organizing source files, and understanding how one class can interact with another.

I also improved my debugging and problem-solving skills. Instead of only looking for a solution, I started checking the error message and project structure to find the actual cause of the problem.

6. Understanding of Object-Oriented Programming

The tutorials gave me a practical understanding of classes, objects, methods, and constructors. I then applied inheritance and polymorphism in later tutorials, which helped me see how related classes can share common features while providing different behaviors.

The practical examples made OOP easier to understand because I could see the concepts working in actual Java programs. I now have a clearer idea of why OOP is useful for organizing and maintaining larger programs.

7. Future Learning Plans

I plan to continue improving my Java skills by building larger applications and practicing OOP concepts through my own projects. I would also like to learn more about databases, Java frameworks, and application development.

I also plan to continue using Git and GitHub so that I can keep my future projects organized and build a stronger programming portfolio.

8. Conclusion

Overall, these tutorials gave me a stronger foundation in Java and a clearer understanding of Object-Oriented Programming. The progression from simple Java files to inheritance, polymorphism, file handling, and a GUI application helped me see how the different concepts can be applied in real programs.

The problems I faced also became part of the learning process. Solving Tutorial 6 error and learning to manage the project with Git and GitHub improved my confidence and problem-solving skills. I believe this experience provides a useful foundation for more advanced programming projects.

9. GitHub Repository URL

GitHub Repository: <https://github.com/M0jz/MUGTABA-ABDALLA-MAMOUN--202505010453--JAVA-.git>